ADEOLU ISRAEL MESOGBORIWON

Process & Automation Engineer | Oil & Gas Production Systems

Ojodu Berger, Lagos, Nigeria | +234 703 717 6627

adeolu.mesogboriwon217@gmail.com | LinkedIn: Adeolu Mesogboriwon

PROFESSIONAL SUMMARY

Chemical Engineering graduate and results-driven process professional with hands-on experience in production optimization, process monitoring, and automation systems within manufacturing and oil & gas environments. Demonstrated ability to improve plant throughput by 120% and reduce equipment downtime by 65% through systematic process control and operational monitoring initiatives. Practical exposure to DCS-based process systems, HAZOP studies, and permit-to-work procedures gained at NPDC (OML 111). Skilled in SCADA/HMI systems, instrumentation, and cross-functional team leadership. Actively developing process simulation competency (HYSYS) to support production optimization and debottlenecking roles in upstream and midstream O&G facilities.

CORE COMPETENCIES

Production Optimization: Process monitoring, throughput improvement, KPI tracking, operational reporting

Process Control & Instrumentation: DCS systems, SCADA/HMI (WinCC), PLC (Siemens S7, AB), control room operations

Automation & Control Systems: PLC programming (Ladder Logic, FBD), VFD drives, motor control, ATS, electro-pneumatic circuits

Field Engineering: Equipment troubleshooting, preventive maintenance, HSE compliance, permit-to-work, HAZOP participation

Data & Reporting: Process data analysis, operational KPI dashboards, Microsoft Excel, production logs

Design Tools: AutoCAD Mechanical & Electrical, Siemens TIA Portal, WinCC SCADA

Standards & Safety: HSE Level 1/2/3, API/ASME awareness, safety system integration

PROFESSIONAL EXPERIENCE

Industrial Automation & Process Engineer (Trainee) *Aug 2025 – Mar 2026*

Institute of Industrial Technology, Isheri-North, Lagos

- Designed and implemented HMI/PLC-based tank level control system, integrating level sensors and control logic to automate pump operation — directly analogous to separator level control in production facilities.
- Configured WinCC SCADA system for real-time process visualization, alarm management, trending, and historical data logging — skills transferable to production monitoring and facility performance tracking.
- Automated conveyor sorting system using proximity sensors and PLC logic, improving operational efficiency by 40% and reducing manual intervention — demonstrating process debottlenecking methodology.
- Implemented VFD-based variable speed motor control, achieving 25% energy efficiency improvement — applicable to compressor and pump optimization in gas handling systems.
- Developed electro-pneumatic and hydraulic control circuits using solenoid valves, DCVs, FCVs, and actuators — relevant to process valve control and flow assurance applications.
- Designed Automatic Transfer Switch (ATS) for seamless grid/generator switching — supporting utilities system reliability knowledge required for facility operability studies.

Production Supervisor *Jul 2023 – Nov 2024*

Industrial Projects International Ltd, Abeokuta

- Improved plant production output by 120% through real-time process monitoring, control optimization, and data-driven decision-making — directly demonstrating production optimization impact.
- Reduced equipment downtime by 65%, saving approximately ₦2.5M/month in lost production costs, through structured preventive maintenance protocols and debottlenecking of production constraints.
- Troubleshot and resolved faults in motors, VFD drives, conveyor systems, and process equipment, achieving 95% first-time fix rate — evidencing systematic constraint identification and resolution.
- Monitored and reported production KPIs, prepared operational reports, and maintained process performance tracking framework across shift operations.
- Supervised and trained a cross-functional team of 30 operators and technicians on equipment operation, safety procedures, and process improvement initiatives.
- Maintained zero lost-time accidents record across all supervised operations, demonstrating commitment to HSE compliance.

Production Intern (Process & Instrumentation) *Jul 2018 – Dec 2018*

Nigerian Petroleum Development Company (NPDC), Benin City

- Monitored daily production parameters and process variables through DCS-based control systems in an active oil & gas production facility — direct exposure to upstream O&G process systems.
- Participated in HAZOP studies, hazard identification reviews, and permit-to-work procedures, gaining foundational knowledge of formal safety engineering processes.
- Supported equipment inspections, shutdown preparation, and commissioning activities following industry best practices and regulatory standards.
- Assisted in oil and water sample analysis, maintaining production logs and daily operational records in compliance with regulatory requirements.

- Gained hands-on exposure to DCS process systems, control room operations, and SCADA monitoring in a live O&G production environment.

EDUCATION

Diploma in Mechatronics Engineering *Mar 2026*

Institute of Industrial Technology, Isheri-North, Lagos

Specialisation: PLC programming, HMI/SCADA systems, automation design, electro-pneumatic/hydraulic control, instrumentation.

B.Eng. Chemical Engineering *Nov 2019*

Federal University of Petroleum Resources (FUPRE), Effurun-Warri

Relevant coursework: Process design, thermodynamics, fluid mechanics, heat & mass transfer, process control, oil & gas production systems.

Data Analysis & Visualisation Certificate *Nov 2024*

TedPrime Hub, Abeokuta — NITDA 3MMT Programme, Cohort 2 (FGN-backed)

CERTIFICATIONS

- HSE Level 1, 2 & 3 — Health, Safety & Environment Certification
- Health, Safety & Basic First Aid Certification
- Data Analysis Certification — NITDA / TedPrime Hub (Nov 2024)
- COREN registration — in progress

KEY TECHNICAL PROJECTS

PLC-Based Automated Tank Level Control System

- Developed complete automation solution integrating level sensors, PLC programming, and HMI interface for industrial tank management — directly analogous to separator and vessel level control in production facilities.

- Implemented fail-safe logic to prevent overflow and dry-run conditions, ensuring continuous operation and equipment protection.

Industrial HMI/SCADA Visualisation System

- Configured WinCC-based SCADA system for real-time process monitoring, data acquisition, alarm management, trending, and historical logging — mirrors production performance tracking requirements in O&G facilities.

Smart Conveyor Sorting & Debottlenecking Study

- Designed and programmed automated sorting system using proximity sensors, pneumatic actuators, and PLC logic, reducing sorting time by 40% through systematic constraint identification and process redesign.

PROFESSIONAL DEVELOPMENT ROADMAP

Actively pursuing the following to close remaining gaps for senior process engineering roles:

- Process simulation proficiency: currently self-studying HYSYS; targeting formal Aspen Technology training within 12 months.
- COREN registration: initiating application process; documentation of engineering experience underway.
- P&ID / PFD development: pursuing short course in process engineering design deliverables.
- Industrial communication protocols: targeting Modbus, HART, and Profibus certification.

REFERENCES

Available on request.